

VEDA TECHNOLOGY — TASK 22

Cohort Retention Basics

Data Analytics Track | Online Retail II

1. Objective

Build a cohort retention table by signup/first-purchase month and use it to understand how customer activity changes over time.

2. Dataset

The project uses the UCI Online Retail II dataset. UCI describes it as a two-year transaction dataset for a UK-based registered non-store online retailer, with 1,067,371 transaction records and eight fields: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID and Country.

3. Data Preparation

Action	Purpose
Remove missing customer/date/quantity/price	Keep records suitable for customer-level retention analysis
Remove cancellation invoices	Exclude InvoiceNo values beginning with C
Keep Quantity > 0 and UnitPrice > 0	Focus on positive completed purchases
Remove exact duplicates	Avoid double counting
Revenue = Quantity × UnitPrice	Create a transaction value field

4. Cohort Method

Each customer is assigned to the month of their first valid purchase. For every later activity month, the script calculates the number of months elapsed since that cohort month. Retention is active customers from the original cohort divided by the original cohort size.

$\text{Retention \%} = \text{Active customers in cohort-period} \div \text{Original cohort customers} \times 100$

5. Deliverables

- Cohort retention table
- Cohort retention heatmap
- Retention curves
- Cohort sizes
- Monthly activity table
- SQL analysis
- Power BI dashboard guide and DAX measures
- Business insights

6. Power BI Dashboard

Recommended visuals are KPI cards, a cohort retention matrix/heatmap, retention curves by cohort, monthly active customer trends, and slicers for Country and Cohort Month.

7. Important Analytical Note

Recent cohorts have fewer observable months. A recent cohort should not be compared with an older cohort at M12 if it has not existed for 12 months. Compare cohorts at the same period number.

8. Execution

Run `cohort_retention_analysis.py`. The script downloads the official UCI workbook when it is not present, cleans the data, creates the cohort tables, writes CSV outputs, and creates the heatmap and retention-curve images.

This report intentionally does not invent cohort percentages. The exact numerical retention results are generated from the official dataset when the script is executed.